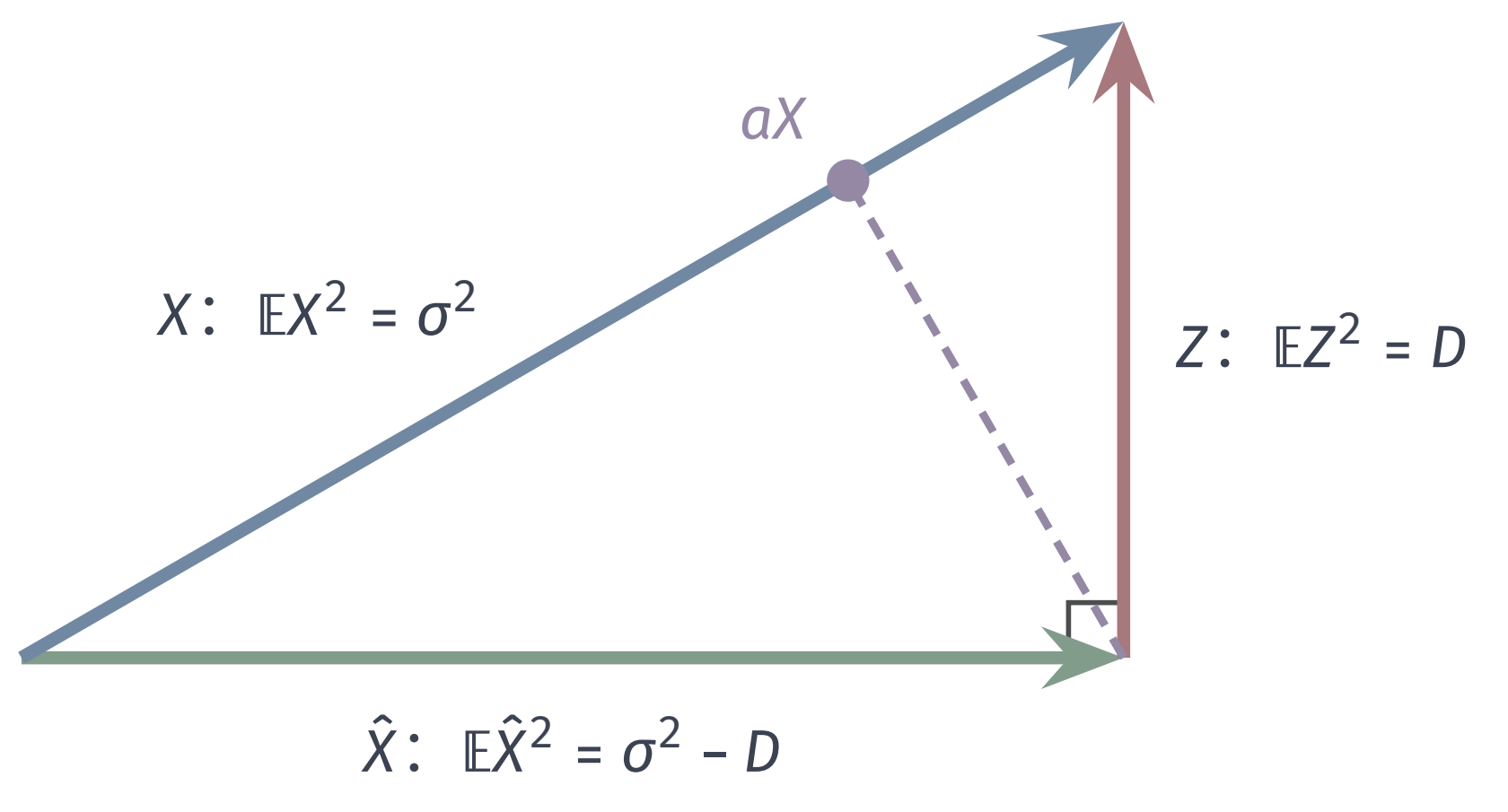
aX
X: $\mathbb{E}X^2 = \sigma^2$
Z: $\mathbb{E}Z^2 = D$
$\hat{X}$: $\mathbb{E}\hat{X}^2 = \sigma^2 - D$